

Assignment 2

Author: Jing Zhang

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Preliminaries

Package Loading

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 3.2.5
```

```
library(haven)
```

```
## Warning: package 'haven' was built under R version 3.2.5
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 3.2.5
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

Assignment Part I

1) *Answer to Problem 1*

5050.

For the sum of natural numbers

$$S_n = \frac{n(n+1)}{2}$$

Since $n=100$ in this case, $S_n=5050$

2) *Answer to Problem 2*

- (a) This function is continuous at $x=0$ since $\lim_{x \rightarrow 0} f(x) = f(0) = 0$
- (b) This function is continuous at $x=1$ since $\lim_{x \rightarrow 1} f(x) = f(1) = 1$
- (c) This function is not a continuous function since this function is not continuous at $x=2$.

3) **Answer to Problem 3**

- (a) $(4x - 2)x^3 = 4x^4 - 2x^3$
 $\frac{\partial 4x^4 - 2x^3}{\partial x} = 16x^3 - 6x^2$
- (b) $x(x + 1)(x + 2) = x^3 + 3x^2 + 2x$
 $\frac{\partial x^3 + 3x^2 + 2x}{\partial x} = 3x^2 + 6x + 2$
- (c) $y = \frac{3x^2 + 1}{2x} = \frac{3}{2}x + \frac{1}{2}x^{-1}$
 $\frac{\partial y}{\partial x} = \frac{3}{2} - \frac{1}{2}x^{-2}$
- (d) $\frac{(2x-1)(x+1)}{x} = 2x + 1 - x^{-1}$
 $\frac{\partial (2x+1-x^{-1})}{\partial x} = 2 + x^{-2}$
- (e) $(x + \frac{a}{x})(x - \frac{a}{x}) = x^2 - a^2x^{-2}$
 $\frac{\partial (x^2 - a^2x^{-2})}{\partial x} = 2x + 2a^2x^{-3}$
- (f) $y = \frac{x-a}{bx^3} = \frac{1}{b}x^{-2} - \frac{a}{b}x^{-3}$
 $\frac{\partial y}{\partial x} = -\frac{2}{b}x^{-3} + \frac{3a}{b}x^{-4}$

4) **Answer to Problem 4**

- (a) $f'(x) = 5x^4 + 3 \geq 3 \forall x \in R$ The function is up-ward sloping and strictly increasing.
- (b) Yes. $f(1) = -8 < 0$; $f(2) = 20 > 0$

According to the intermediate value theorem, there is at least one real number $1 < x < 2$ that $f(x) = 0$

- (c) Yes. $f(1) < 0$; $f(\frac{3}{2}) > 0$.

According to the intermediate value theorem, there is at least one real number $1 < x < \frac{3}{2}$ that $f(x) = 0$

- (d) No. According to (a), we know that the function is upward sloping, means that there is at most one solution that make $f(x) = 0$, we already know that such x is bigger than 1 and less than $\frac{3}{2}$, so there is no x between $\frac{3}{2}$ and 2 that makes $f(x) = 0$.

Assignment Part II

5) **Answer to Problem 5**

The country that experiences the largest percentage increase in Islam is Qatar, and the percentage increase is 0.2737.

```
religions<-read_dta("~/Dropbox/courses/Mathematic Foundations of Political Methodology/Weekly R lab/week
religionchange <- religions %>%
group_by(country) %>%
summarise(maxinc = max(c(diff(isgenpct), 0), na.rm = T))

religions<-religions%>%left_join(religionchange)
```

```
## Joining, by = "country"
```

```
which.max(religions$maxinc)
```

```
## [1] 1566
```

```
religionsmax<-select(religions, year, country,isgenpct, maxinc)
religionsmax[1566, ]
```

```
## # A tibble: 1 × 4
##   year country isgenpct maxinc
##   <chr>   <chr>     <dbl>  <dbl>
## 1  1970   Qatar         1 0.2737
```

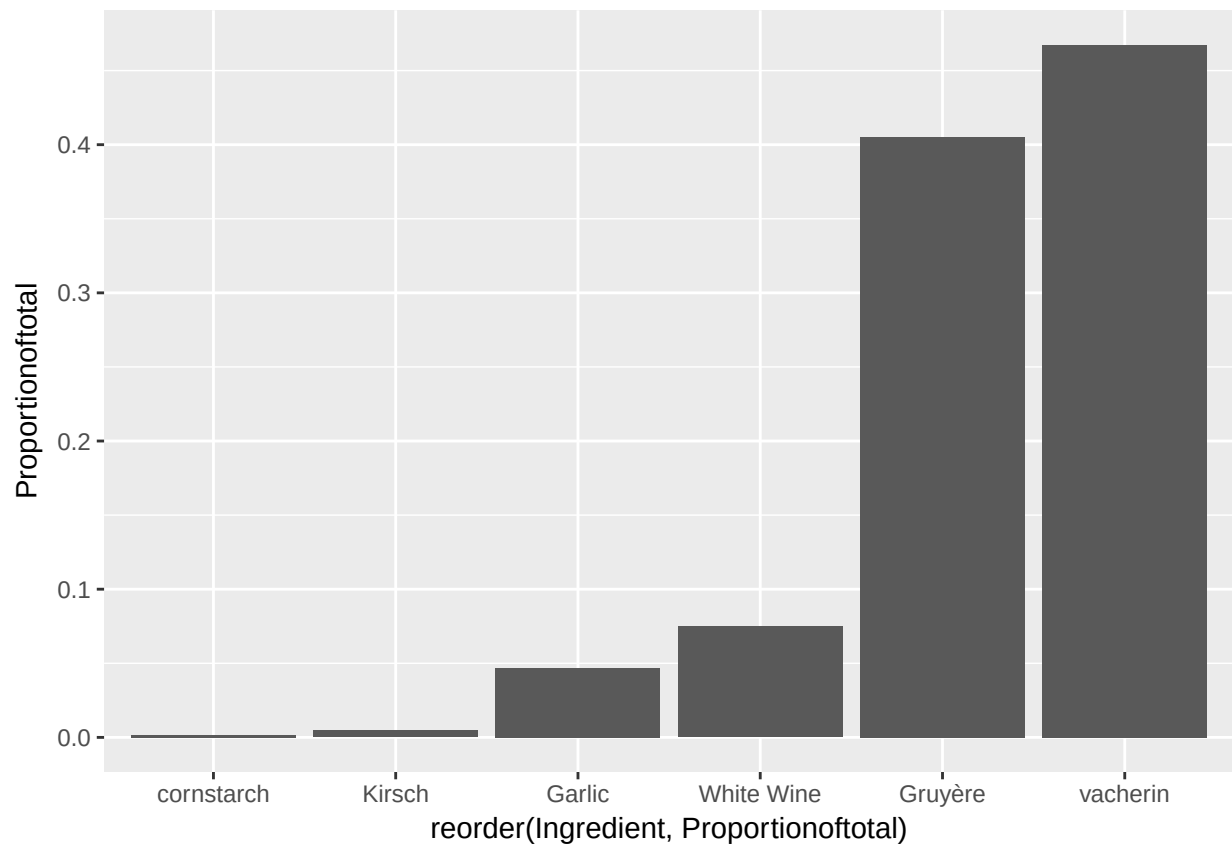
6) *Answer to Problem 6*

- (a) df1: 11 rows (0:10) and 3 columns (x, yA, yB).
- (b) df2: 6 rows (5:10) and 3 columns (x, yA, yB). Keep rows that both dfA and dfB has values.
- (c) df3: 5 rows (0:4) and 2 columns (x, yA).. Return rows that only dfA has values.

7) *Answer to Problem 7*

```
dfFondue <- data.frame(Ingredient=c("Gruyère", "vacherin", "Kirsch","Garlic",
                                   "White Wine", "cornstarch"))%>%
mutate(Origin = c("Gruyères", "Fribourg", "Zürich", "Lucern", "Jura", "Iowa"))%>%
mutate(Price=c(12.99, 14.99, 15.00, 1.50, 12.00, .90))%>%
mutate(Amount=c(1, 1, .01, 1, .2, .05))%>%
mutate(Proportionoftotal = Price*Amount/sum(Price*Amount))

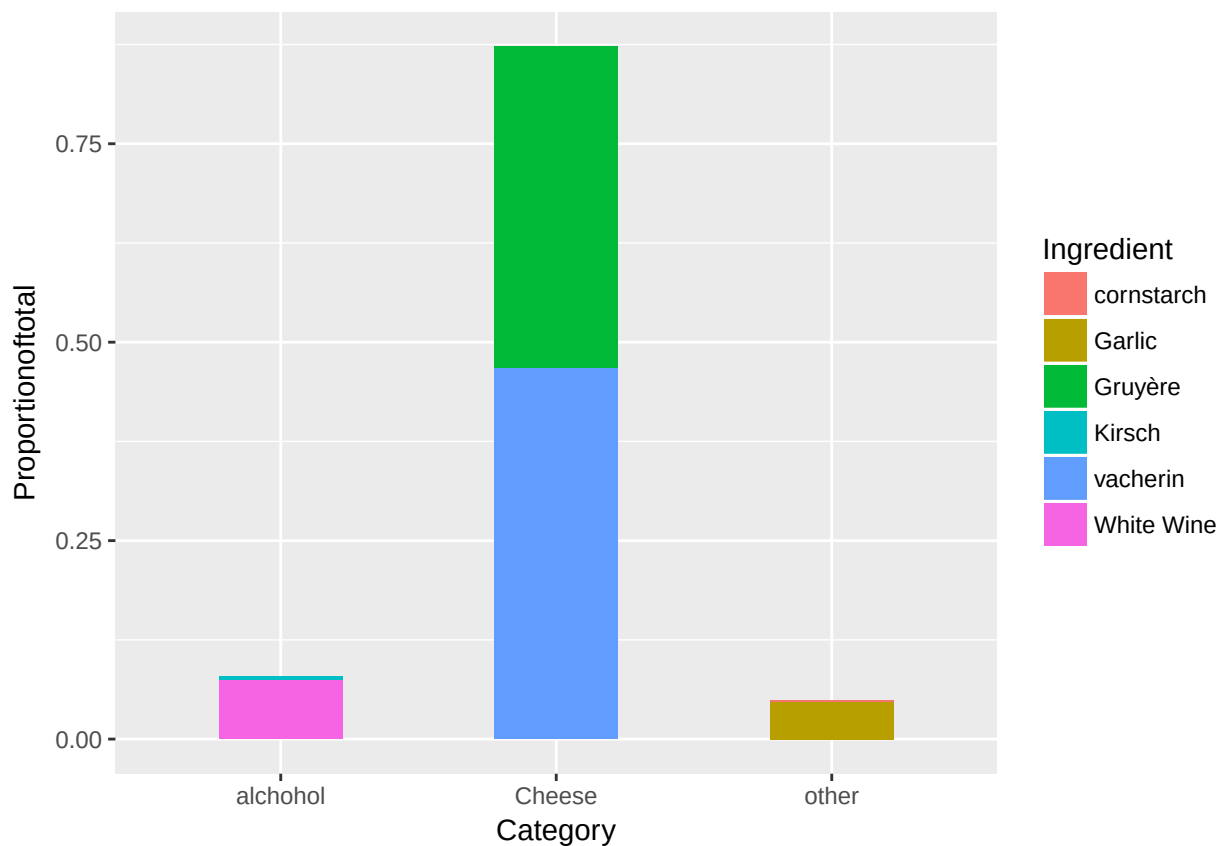
ggplot(dfFondue, aes(reorder(Ingredient,Proportionoftotal), Proportionoftotal))+geom_col()
```



8) *Answer to Problem 8*

```
attach(dfFondue)
dfFondue$Category[Ingredient == "White Wine"] <- "alcohol"
dfFondue$Category[Ingredient == "Garlic"] <- "other"
dfFondue$Category[Ingredient == "cornstarch"] <- "other"
dfFondue$Category[Ingredient == "vacherin"] <- "Cheese"
dfFondue$Category[Ingredient == "Gruyère"] <- "Cheese"
detach(dfFondue)

figure <- ggplot(data=dfFondue, aes(x=Category, y=Proportionoftotal, fill=Ingredient)) +
  geom_bar(stat='identity', position='stack', width=0.45)
figure
```



9) *Answer to Problem 9*

(1) The number of sampled individuals in 1867 that Absconded is 90.

```
ICPSR <- read_dta("~/Dropbox/courses/Mathematic Foundations of Political Methodology/Assignments/Assignment 9/ICPSR.dta")
table(ICPSR$ABSCONDED, ICPSR$YEAR==1867)
```

```
##
##      FALSE  TRUE
## -98 536424 15714
##  0  772263 21393
##  1    942    90
```

(2) Their average age is 46.09804 (keep 5 points decimal)

```
ICPSR%>%  
  group_by(ABSCONDED)%>%  
  summarize(sumAGE=mean(AGE_IN_SUI[YEAR==1867&AGE_IN_SUI!=-99])) %>%  
collect
```

```
## # A tibble: 3 × 2  
##   ABSCONDED    sumAGE  
##   <dbl+lbl>    <dbl>  
## 1      -98 34.59485  
## 2         0 32.40944  
## 3         1 46.09804
```

10) *Answer to Problem 10*

I checked the association between the percentage of adherences in Islam (isgenpct, independent variable) and civil war (onset, dependent variable).

As the median of isgenpct is 0.0194, I coded “high” for isgenpct that is equivalent or higher than 0.0194, and “low” for the others. From the table, we can see that countries with high adherence in Islam are more likely to go into the civil war (0.014) than those of low adherence in Islam (0.012).

However, the association between war onset and percentage of adherence in Islam seems not to be statistically significant as shown from the result of the probit regression.

```
fldta <- read_dta("~/Dropbox/courses/Mathematic Foundations of Political Methodology/Assignments/Assignm")

fldta <- fldta %>%
  select(
    cowcode = ccode,
    year = year,
    cname = cname,
    onset = onset,
    war = war,
    polity2 = polity2,
    Oil = Oil,
    GDPpercap = gdpen
  )
```

```
religions <- religions %>% mutate(year = as.numeric(gsub(",", "", year)))
ReligionWar<- religions%>%full_join(fldta)
```

```
## Joining, by = c("year", "cowcode")
```

```
median(ReligionWar$isgenpct, na.rm=T)
```

```
## [1] 0.0194
```

```
warreligions1 <- mutate (ReligionWar, religiousity=ifelse(isgenpct>=0.0194, "high", "low"))
round(prop.table(table(warreligions1$religiosity, warreligions1$onset), margin=1),3)
```

```
##
##           0      1      4
##   high 0.986 0.014 0.000
##   low  0.988 0.012 0.000
```

```
probit <- glm(onset ~ isgenpct, data=ReligionWar, family=binomial(link="probit"))
summary(probit)
```

```
##
## Call:
## glm(formula = onset ~ isgenpct, family = binomial(link = "probit"),
##      data = ReligionWar)
##
```

```

## Deviance Residuals:
##      Min        1Q      Median        3Q        Max
## -0.2004   -0.1666   -0.1530   -0.1523    2.9877
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.2726     0.1175  -19.34  <2e-16 ***
## isgenpct      0.2164     0.2430    0.89   0.373
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 180.46  on 1270  degrees of freedom
## Residual deviance: 179.71  on 1269  degrees of freedom
## (6063 observations deleted due to missingness)
## AIC: 183.71
##
## Number of Fisher Scoring iterations: 7

```